

Curriculum Vitae – Aidan J Carlson

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EDUCATION

Ph.D. Student **Michigan Technological University**, Houghton, MI
Aug 2026 – Present Mathematical Sciences – Applied & Computational Mathematics

Bachelor of Science, **University of Minnesota Duluth**, Duluth, MN
Aug 2022 – May 2026 Cum Laude with Honors, GPA: 3.885
Three Majors: Mathematics, Actuarial Science and Statistics,
Computer Science

PROFESSIONAL EXPERIENCE

Michigan Technological University, *Department of Mathematical Sciences*
Aug 2026 - Present **Graduate Teaching Assistant**
Courses: MA 1161 (Calculus Plus with Technology I)

University of Minnesota Duluth (UMD), *Department of Mathematics and Statistics*
Jan 2025 – May 2026 **Undergraduate Grader**
Courses: MATH 3326 (Vectors and Matrices),
MATH 4326 (Linear Algebra)

Jan 2023 – May 2026 **Undergraduate Teaching Assistant**
Courses: MATH 103 (Basic Math), MATH 1005 (College Algebra),
MATH 1250 (Precalculus)

University of Minnesota Duluth, *Department of Advising/ Academic Services SCSE*
Jan 2024 – May 2026 **Learning Assistant**
Courses: MATH 1160 (Finite Math and Introduction to Calculus),
MATH 1296 (Calculus I), MATH 3298 (Calculus III)

AWARDS / HONORS

May 2026 UMD Mathematical Sciences Outstanding Senior
May 2026 UMD Math & Stats Departmental Honors
May 2026 UMD Computer Science Departmental Honors
2025 – 2026 UMD Sylvan D. Burgstahler Memorial Scholarship (\$2,307)
2022 – 2026 UMD Scholarship (\$10,000)

2025 – 2026	UMD James L. Nelson Memorial Scholarship (\$2,000)
Fall 2023	Undergraduate Research Opportunities Program (\$1,800)
June 2022	Eagle Scout, BSA

SCHOLARSHIP

Refereed Articles

1. **Carlson, A.**, Froncek, D., **(2026)** *Decompositions of Complete Graphs into Disconnected Unicyclic Triangular Graphs with Seven Edges*, The Australasian Journal of Combinatorics, SUBMITTED
2. Banegas, D., **Carlson, A.**, Hawkes, L., Heck, M., Higgins, C., Hong, Y., Jiang, K., Palme, N., Qi, D., Sundvall, X., Waadevig, J., Freyberg, B., Froncek, D., **(2026)** *Decomposition of Complete Graphs on $n \equiv 0, 1 \pmod{14}$ Vertices into Disconnected Unicyclic Tripartite Graphs on Seven Edges*, J. Combin. Math. Combin. Comput. 129, 63-67
3. Banegas, D., **Carlson, A.**, Froncek, D., **(2026)** *Decomposition of Complete Graphs into Disconnected Unicyclic Graphs with Seven Edges and More Than Eight Vertices*, J. Combin. Math. Combin. Comput. 129, 79-94
4. Bevers, M., **Carlson, A.**, Pedersen T., **(2026)** *Duluth at SemEval-2026 Task 4: A Hybrid Approach to Narrative Similarity Using Bi-Encoder Embeddings with Cross-Encoder Tie Breaking Using Learned Weights*. In *Proceedings of the 20th International Workshop on Semantic Evaluation*, 927-931, San Diego, CA, USA. Association for Computational Linguistics
5. **Carlson, A.**, Hopkins, B., Sellers, J. A., **(2024)** *Enumeration Modulo 4 of Overpartitions Wherein Only Even Parts May Be Overlined*, Discrete Mathematics Letters 14, 95-102

In Progress

Carlson, A., Hampton, M., Jenson, A., *Finiteness of Planar 4-Body Central Configurations with Homogeneous Potential*

Presentations

2026 Undergraduate Research and Artistic Showcase, University of Minnesota Duluth (April 28th, 2026) Title: *A Hybrid Approach to Narrative Similarity using Bi-Encoder Embeddings with Cross-Encoder Tie Breaking and Learned Weight*

2026 57th Southeastern International Conference on Combinatorics, Graph Theory & Computing, Florida Atlantic University (March 9th-13th, 2026) Title: *Decompositions of Complete Graphs into Disconnected Triangular Graphs with 7 Edges*

2024 35th Midwestern Conference on Combinatorics and Combinatorial Computing, University of Minnesota Duluth (October 18th – 20th, 2024) Title: *Decompositions of Complete Graphs into Certain Unicyclic Bipartite Graphs with Seven Edges*

2024 Spring Meeting of the North Central Section of the MAA, University of St. Thomas (April 6th, 2024) Title: *Overpartitions with Only Even Parts Overlined*

2023 Undergraduate Colloquium, University of Minnesota Duluth (November 16th, 2023) Title: *Overpartitions with Only Even Parts Overlined*

PROGRAMMING SKILLS

Languages:	C, C++, C#, Dart, Java, JavaScript, LaTeX, Python
Frameworks:	React.js, Node.js, Express.js, Flutter
Databases:	PostgreSQL, MySQL, Firebase, MongoDB

SERVICE

2025 Reviewer for the Institute of Electrical and Electronics Engineers, International Workshop on Quality of Service (IEEE IWQoS) (Reviewed one paper/ Operating Systems)

2024 Reviewer for the Minnesota Undergraduate Research and Academic Journal (MURAJ) (Three month course on peer reviewing fundamentals, reviewed one paper/ Statistics)